

Neural Adaptations to Endurance Training

4

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Neural adaptations induced by strength training have been widely described although recent technical developments (e.g., transcranial magnetic stimulation) have provided new insights. Neural adaptations to endurance training are not as well-known and usually considered to be much smaller than those observed following strength training. In this chapter, we will not use the real definition of endurance, that is the ability to sustain a high percentage of maximal oxygen uptake ($\text{VO}_{2\text{max}}$). Instead, we will use common usage of the word endurance, i.e., prolonged, low-intensity exercise, usually with large muscle mass such as cycling, running, and cross-country skiing. The theory behind chronic adaptations is related to acute deleterious effects and recovery. In the first part of this chapter, the tools used to assess neural adaptations will be briefly described. Then, we will focus on the acute neural responses induced by a single endurance training session. Special consideration will be given to the difference between endurance running and cycling/cross-country skiing at the end of this second section. The third part of this chapter will be dedicated to chronic adaptations to the neural command induced by endurance training.

Tools Used to Assess Neural Adaptations

The tools used to assess neural adaptations are described in Fig. 4.1. The outcome measures used to assess neuromuscular fatigue and training adaptations are force and (usually surface) electromyographic activity (EMG). EMG involves the placement of electrodes on the muscle(s) of interest (agonist and antagonist muscles) to measure electrical activity (i.e., sum of the action potentials) within the muscles. Force is

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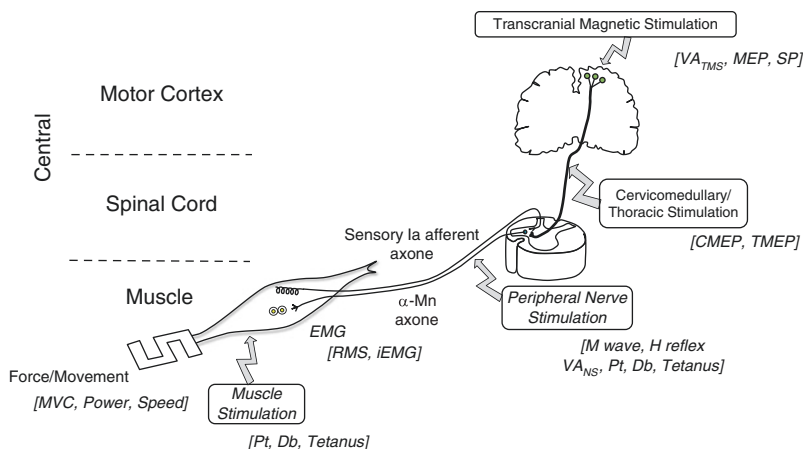


Fig. 4.1 Neuromuscular function: how to measure it? Adapted from [3]. The main outcomes are presented in *italics*. *CMEP* cervicomedullary motor-evoked potential, *Db* doublet, *H-reflex* Hoffmann reflex, *iEMG* integrated EMG, *MEP* motor-evoked potential, *MVC* maximal voluntary contraction, *M wave* compound muscle action potential, *NS* nerve stimulation, *Pt* peak twitch, *RMS* root mean square, *SP* silent period, *TMEP* thoracic motor-evoked potential, *TMS* transcranial magnetic stimulation, *VA* maximal voluntary activation

generally assessed isometrically via force transducer in line with the applied force. The gold standard for assessing central drive and central fatigue development is voluntary activation (VA) measured with the interpolated twitch technique. During a maximal voluntary contraction (MVC), a supramaximal stimulus is delivered to the peripheral nerve or muscle at maximal voluntary force and the resulting superimposed twitch (i.e., additional force elicited by the stimulus) is compared to a twitch (the evoked mechanical response to a supramaximal stimulus) elicited by supramaximal stimulation immediately afterwards in the relaxed muscle [4]. This provides a comparison of the force produced by recruiting all unrecruited motor units (MUs) (all MUs are theoretically recruited during MVCs) and the maximal force capacity by simultaneously recruiting all MUs in the muscle. While this method only quantifies fatigue for the central component as a whole (i.e., central nervous system including the brain and motoneurons), the assessment of VA using TMS provides information on supraspinal fatigue [5]. A methodological difference when employing TMS to determine VA is that the resting twitch must be estimated from superimposed twitches from a series of contractions between 50 and 100% of MVC because during the transition from rest to light/moderate-intensity voluntary contractions, the corticospinal pathway excitability increases rapidly. Other methods for assessing central drive to the muscles utilize EMG. The overall electrical activity of the muscle (i.e., central drive) is usually expressed as root mean square (RMS) or integrated EMG (iEMG). Supramaximal stimulation of the motor nerve during strong or maximal voluntary contractions can also be used to elicit V waves. V waves are the result of collisions between evoked antidromic and voluntary orthodromic volleys and the

resulting reflex response. Only axons actively involved in the voluntary contraction will contribute to the subsequent EMG response [6].

Motoneuronal excitability can be assessed by submaximal stimulation of the motor nerve in a relaxed or lightly contracting muscle to elicit an afferent volley before recruiting motoneurons. The resulting EMG response is referred to as the Hoffmann reflex (H-reflex). While indicative of motoneuronal excitability, use of this method is limited to certain muscles (e.g., soleus, SOL) and is influenced by presynaptic inhibition [6]. Increasingly common is the use of spinal stimulation via direct (usually electrical) stimulation of the descending tracts in the spinal cord. Providing information on the excitability of the motoneuron pool, transmastoid stimulation elicits cervicomedullary motor-evoked potentials (CMEP) in upper-limb muscles that are recorded as EMG [7]. Due to difficulties in eliciting CMEPs in the lower limbs, spinal stimulation is generally performed at the upper-thoracic level to elicit thoracic motor-evoked potentials (TMEP) [6].

The investigation of cortical excitability employs transcranial magnetic stimulation (TMS) to deliver a magnetic pulse to the motor cortex that trans-synaptically excites pyramidal axons to elicit a response that is transmitted via the motoneurons to the muscle and recorded in EMG as a motor-evoked potential (MEP) [8]. Since the MEP is indicative of the excitability of the entire corticospinal pathway, it must be normalized to a corresponding CMEP or TMEP elicited at the same time and under the same conditions to isolate cortical excitability. Another TMS-elicited parameter, the silent period (SP), is commonly regarded as indicative of intracortical inhibition, although the first 150 ms are mainly influenced by spinal mechanisms and, thereafter, by cortical mechanisms [9].

Since EMG is recorded at the muscle level, methods using EMG responses to quantify central fatigue and changes in motoneuronal and corticospinal excitability must take into account any changes that occur within the muscle. Supramaximal stimulation of the motor nerve elicits a compound muscle action potential (M wave, Mmax) in EMG from the efferent volley, indicative of sarcolemmal propagation within the muscle. M waves elicited at the same time and in the same muscle state are used to normalize EMG responses to account for muscle changes.

However, it is important to recognize the limits of the aforementioned methods for assessing neural adaptations and acute changes. Methods such as the V-wave and H-reflex are performed in very specific conditions that may not always be applicable to certain situations or applications [6]. Observed changes in MEPs and SPs in fatigued conditions may also be influenced by TMS stimulus intensity alone [10] and, as observed in later sections, may result in conflicting responses for the same exercise bout. Methodological advances also continue to be made so that investigations can more accurately capture both the type and magnitude of fatigue. One example of a recent methodological development is a new cycling ergometer permitting dynamic exercise and subsequent isometric neuromuscular evaluations within ~1 s after the end of exercise [11]. This new ergometer eliminates recovery prior to post-exercise evaluations, an important development since fatigue-related neuromuscular changes recover quickly, sometimes within seconds [12–14].

Acute Neural Responses Induced by a Single Endurance Training Session: Central Fatigue

Definition of Fatigue and Its Central Component

Fatigue is the reduction in the capacity to voluntarily produce force (e.g., [15]). Fatigue can be further categorized as being either peripheral or central. Peripheral fatigue occurs distal to the neuromuscular junction. Central fatigue occurs within the central nervous system and results in a decrease in maximal neural drive to the muscle. This most commonly presents as a decrease in the ability to voluntarily contract a muscle maximally and the presenting fatigue can be further identified as being within the entire central component (i.e., from the brain and motoneurons) or solely at the supraspinal level (i.e., within the brain), depending on the technique used (see section “Tools Used to Assess Neural Adaptations” on methods to assess fatigue).

Central Fatigue Induced by Endurance Exercise

Numerous studies have explored the etiology of acute fatiguing endurance exercise (i.e., a single exercise bout). Central fatigue has been observed following acute exercise bouts, most specifically cycling and running. As such, this chapter focuses solely on the knee-extensor and plantar-flexor muscle groups since these muscle groups pertain directly to cycling and running. In the knee-extensor muscles, the magnitude of supraspinal fatigue (i.e., decrease in VA_{TMS}) increases as exercise duration increases, from approximately 8–11% for exercise lasting several hours [16, 17] to 16% for exercise longer than 13 h [10]. One exception is Klass et al. [18], who did not observe any change in VA_{TMS} after 60 min of cycling at 55% of maximal aerobic power (W_{max}) followed by a ~30-min time-trial; however, VA_{TMS} was measured 10 min after the cycling exercise terminated. The magnitude of knee-extensor central fatigue increases from <20% for exercise lasting up to 5 h (e.g., [16, 19–22]) to >25% for exercise longer than 13 h [10, 20, 21]. The magnitude of central fatigue observed in the plantar flexors is less than for the knee extensors and ranges from 6 to 15% for exercise longer than 13 h [20, 21, 23]. The inhibitory effect of group III and IV afferents has been highlighted as a major contributing factor to voluntary activation impairment following fatiguing cycling exercise [16, 24].

Potential changes to the corticospinal pathway, as measured by EMG responses, have also been observed. There may be increases [10, 16] or no change [10, 16–18, 25] in knee-extensor corticospinal excitability (i.e., MEP size) following a prolonged acute exercise bout. Decreased motoneuronal excitability was observed following 90 min of running as demonstrated by a decreased magnitude of both V-wave and H-reflex responses in the plantar flexors [26]. Conversely, motoneuronal excitability as indicated by CMEP area was unchanged following shorter duration (~8–9 min), high-intensity cycling bouts [27, 28]. There may also be increases [10] or no change [10, 16, 18, 24, 29] in knee-extensor SP duration following an acute

cycling bout. One limitation of all of these studies is that the delay to post-exercise evaluations permitted recovery to occur, meaning that the amount of central fatigue and changes to the corticospinal pathway were very likely underestimated. As described above, recent methodological developments have allowed this limitation to be overcome.

Does the Type of Endurance Activity Impact Central Fatigue?

The response to this question is a definite yes. Even though a direct comparison with the same subjects exercising for the same duration and at the same intensity still needs to be conducted, we systematically found that running induced more central fatigue than either cycling or cross-country skiing, two activities that induce less mechanical stress/fatigue in the muscles [30, 31]. This was the case when comparing 5-h running and cycling bouts at 55% of $\text{VO}_{2\text{max}}$ [19, 22] and a ski-skating marathon vs a 30-km trail-running competition [15, 32]. These results suggest that neurobiological changes such as the accumulation of serotonin, play only a moderate role, if any, in central fatigue development during prolonged exercise. The potential inhibitory action from thin afferent fiber (group III–IV) signaling, which may be sensitized by pro-inflammatory mediator production during prolonged running exercise, may explain the large central fatigue after ultra-marathons [20, 21, 33]. Yet, indirect evidence showing that a large inflammatory response persisted 2 days after a 165-km mountain ultra-marathon while VA had returned to pre-race values [21] does not support this hypothesis. Another possible explanation for the difference between the magnitude of fatigue elicited by running and cycling is that the disfacilitation of group Ia afferents may be more pronounced in running than cycling due to repeated eccentric muscle actions. This could induce fatigue/damage at the spindle level or simply increase muscle compliance [34].

Neural Adaptations Induced by Chronic Systematic Endurance Training

It is well-known that endurance training improves fitness. Mechanisms such as increased stroke volume, partly due to greater blood volume, and increased muscle oxidative capacity (e.g., mitochondrial content) have been proposed to explain the improvements in aerobic fitness (e.g., [35, 36]). It is also well-known that strength training, particularly explosive and plyometric training, can trigger neural adaptations (e.g., [37, 38]). For instance, using a combination of V-wave, H-reflex, and Mmax responses, Aagaard et al. [37] demonstrated that 14 weeks of heavy-weight resistance training induced an increase in maximal central drive, likely mediated by a combination of both supraspinal and spinal adaptations. Earlier MU activation, an increase in the number of doublets and triplets and an enhanced maximal discharge rate were observed after 12 weeks of dynamic training (at 30–40% of maximal force) [39]. Changes in corticospinal excitability were also evidenced after resistance

training (see below). Here, we aim to describe the lesser known neural adaptations after endurance training. Although adaptations within the nervous system are not the main reason patients, sedentary people or athletes perform endurance exercise, it is important to understand the potential benefits or, conversely, the deleterious role that endurance training can have on central drive. We will first compare central nervous system (CNS) outcomes (e.g., VA, H-reflex) in endurance athletes vs untrained subjects and athletes trained in sports that require greater levels of power or strength. Thereafter, we will review the studies that have investigated the effects of an endurance training intervention on the nervous system, at rest and then during and after exercise (i.e., in a fatigued state).

Comparison of CNS Outcomes in Endurance Athletes Versus Explosive Athletes and Sedentary People

One way to examine training adaptations is to compare populations with different training backgrounds. This method has obvious flaws (e.g., impact of genetics, uncontrolled type, and amount of training) but it also has one main advantage: it allows examination of training adaptations due to several years of training as opposed to several weeks (generally 3–12) in interventional studies, i.e., a very limited time period compared to an athlete's or patient's life. Lattier et al. [40] showed that although power-trained athletes (POW) performed much better for squat jumps than endurance-trained athletes (END), the two groups were equally strong for isometric MVC. Also, both groups were stronger than sedentary subjects (SED). The difference might be due to a significantly lower VA in SED while VA was similar between END and POW. The results of Lattier et al. [40] were confirmed by Garrandes et al. [41], who did not find any difference in knee-extensor VA, measured with the interpolated twitch technique, between END and POW. This suggests that both POW and END athletes are equally capable of maximally driving their muscles when activation is measured in isometric conditions. This is likely no longer true when a much higher discharge rate is needed during explosive movements. However, these results have been challenged by Cohen et al. [42], who observed no difference in isometric MVC between END and SED. Although VA was not reported in this study, it is very unlikely that VA was lower in SED than in END. However, both Lattier et al. and Cohen et al. [40, 42] observed a lack of difference in squat jump and rate of force development (RFD) for SED and END. It has also been shown that even when force is expressed as a percentage of maximum, the force-time curves of POW were faster than END. While the amount of neural activation can partly explain this result, these data do not rule out the qualitative characteristics of the muscle tissue itself [43]. Interestingly, RFD was not lower in END than in strength-trained athletes.

In agreement with other studies (see [44]), Maffiuletti et al. [45] showed that the SOL H-reflex normalized to Mmax was significantly smaller in POW than END, suggesting increased α -motoneuronal excitability after endurance training. However, POW exhibited a greater force-to-EMG ratio, meaning POW produced greater force

per number of MUs activated [46]. While the larger H-reflex may simply reflect a greater percentage of type I fibers, presynaptic inhibition, through activation of GABA_A and/or GABA_B receptors of Ia afferent terminals, may be greater for END than either POW or SED [47]. Likewise, it has been shown that stretch reflexes [48, 49] were different depending on training background, possibly due to a higher percentage of type I fiber muscle-spindle density in END.

Training Interventions: Effects on CNS at Rest

Endurance training has been shown to enhance cognitive and neural plasticity in several brain regions including the cerebellum, hippocampus, and cerebral cortex and angiogenesis in the motor cortex [48]. Here, we will only examine the effects of endurance training on motor cortical and corticospinal factors, mainly VA, MEP, and SP as well as V-wave and H-reflex measured at rest, i.e., without fatigue. Data are summarized in Table 4.1.

Unlike the vast majority of strength training programs, MVC did not change in most (e.g., [25, 44, 50, 52, 54]) endurance training interventions. MVC was found to increase in one study [53] but the training was of low-force concentric and isometric muscle actions, i.e., not a traditional cycling or running intervention. The most global indication of neural adaptation, i.e., VA (PNS or TMS), does not change with endurance (or strength) training but this is likely due to a ceiling effect. Indeed, in healthy subjects lower-limb VA is nearly complete prior to training, thus there is little room for improvement. In addition, VA is known to be a semi-quantitative measure that probably does not allow quantification of small increases in central drive [55]. One must also keep in mind that while dynamic training is employed, tests of voluntary activation (and often MVC) are isometric. In other words, the lack of improvement may be indicative of the highly specific nature of the adaptations [56]. One difference between endurance and strength training that could partly explain the difference in terms of MVC improvements (even short-term training where only neural adaptations occur [44]) could be the increased descending drive that occurs after strength training only. Indeed, while V waves were found to increase by 50–80% after strength training (e.g., [37, 44], and the changes in V waves also correlated to MVC gains [44]), this factor did not change after endurance training [44, 50].

At the spinal level, in line with cross-sectional studies (e.g., [45]), it was initially found that the H-reflex increased with endurance training [44, 51]. For instance, Perot et al. [51] found that H reflexes and tendon-tap reflexes increased in 75% of the participants after 8 weeks of endurance (running) training. It was suggested that endurance training increased both the excitability of the motoneurons and the response of the muscle spindles to stretching. Yet the effect of endurance training on the H-reflex pathway has been recently challenged [50]. Behrens et al. [50] used a better experimental design in the sense that they had a control group. As stated by the authors, a dependent t-test on the Hmax/Mmax ratio from pre- to post-training was significant and based on this result, they could have concluded that endurance

Table 4.1 Effects of endurance training on CNS characteristics at rest

Study	Subjects	Age as mean $\pm$ SD or [range] (year)	Control group	Training duration (weeks)	Number of sessions or duration/week	Training characteristics	Muscle(s) (muscle group)	Main outcomes (at rest)
Behrens et al. [50]	10 F, 12 M	CG: 23 ± 2 TRAIN: 24 ± 2	Y	8	2×1 h	Cycling at 70–90% HRmax	SOL (PF)	$VA_{PNS} \leftrightarrow$ H-reflex at rest $\leftrightarrow$ H-reflex during MVC $\leftrightarrow$ V-wave $\leftrightarrow$ RFD $\leftrightarrow$
O’Leary et al. [25]	2 F, 8 M	27 ± 6	N	6	3	Cycling high-intensity IT ($6-8 \times 5$ min at $\Delta 50\%$)	VL (KE)	$VA_{PNS} \leftrightarrow$ $VA_{TMS} \leftrightarrow$ MEP/Mmax $\leftrightarrow$
O’Leary et al. [25]	2 F, 8 M	27 ± 4	N	6	3	Cycling moderate-intensity continuous training ($\sim 60-80$ min at 90% LT)	VL (KE)	$VA_{PNS} \leftrightarrow$ $VA_{TMS} \leftrightarrow$ MEP/Mmax $\leftrightarrow$
Perot et al. [51]	6 F, 14 M	[19–23]	N	8	2 h	Running at 60% MAV	SOL (PF)	Hmax/Mmax $\uparrow$ T-reflex $\uparrow/\leftrightarrow$
Vila-Cha et al. [52]	18 M	26.1 ± 3.9^a	Y	6	3	Cycling increasing from 20–30 min at 50–60% HRR to 40–50 min at 65–75% HRR	VM, VL (KE)	MU discharge rate at 30% MVC $\downarrow$ MU conduction velocity $\uparrow$
Vila-Cha et al. [44]	2 F, 11 M	24.0 ± 2.6^a	N ^b	3	3	Cycling increasing from 30–40 min at 55–65% HRR to 40–50 min at 65–75% HRR	SOL (PF)	Hmax/Mmax $\uparrow$ V-wave $\leftrightarrow$ H-reflex excitability threshold $\downarrow$

Zghal et al. [53]	23 M	CG: 23.9 ± 2.0 TRAIN: 25.2 ± 2.3	Y	8	3	Low-force concentric and isometric contractions	RF, VL, VM (KE)	VA _{PNS} ↔ VA _{TMS} ↔ MEP ↔ SP ↔
Zghal et al. [54]	16 M	23.6 ± 2.6	Y	8	2–3	Running short IT (80–100% MAV) + continuous (60–65% MAV)	RF, VL, VM (KE)	VA _{PNS} ↔ VA _{TMS} ↔ MEP ↔ SP ↔

↔ not statistically significant, ↑ indicates a statistically significant increase, ↓ indicates a statistically significant decrease, Δ50% halfway between lactate threshold and maximal oxygen uptake, CG control group, F female, H_{max} maximal Hoffmann reflex, HR_{max} maximal heart rate, HRR heart rate reserve, IT interval training, KE knee extensors, LT lactate threshold, M male, MAP maximal aerobic power, MAV maximal aerobic velocity, MEP motor-evoked potential, Mmax M wave amplitude or area, MVC isometric maximal voluntary contraction, MU motor unit, PF plantar flexors, RF rectus femoris, RFD rate of force development, SOL soleus, SP silent period, TRAIN training group, T-reflex T-response after Achilles tendon tap (stretch reflex), VA_{PNS} voluntary activation assessed via peripheral nerve stimulation, VA_{TMS} cortical voluntary activation assessed via transcranial magnetic stimulation, VL vastus lateralis, VM vastus medialis

^aAll subjects, including a group that had performed strength training

^bThis study also included a strength training group

training increased the normalized H-reflex response. However, this was not the case with appropriate statistical analysis (ANCOVA). Alternative explanations for the lack of significant change in the H-reflex response in this latter study [50] were that subjects only trained 2 times/week and training was performed as cycling where SOL muscle activation is relatively low compared to the knee-extensor muscles. This could explain the difference with transversal studies where there has been a higher dose of endurance training for several years. Thus, it is possible that (1) a sustained period of regular aerobic exercise is necessary to induce alterations in H-reflex responses and (2) running training has stronger effects on spinal reflex responses than cycling training [50].

To the best of our knowledge, only four studies have examined CNS outcomes using TMS after endurance training [25, 50, 53, 54]. Although these studies found significant increases in endurance performance after 6–8 weeks of various types of endurance training (running vs cycling, continuous vs interval training, local muscle endurance), none of them found significant changes in VA, MEP, or SP measured in resting conditions after training. The effects of resistance training on corticospinal excitability is unclear as some studies showed reductions or no change [38, 57] or increases [58–60] in MEP responses.

It has been shown that 6 weeks of strength and endurance training elicit similar increases in muscle fiber conduction velocity measured at 30% MVC but opposite changes in MU discharge rates. Indeed, MU discharge rate was found to increase after strength training and decrease after endurance training [52]. It is important to note that this was not tested during MVC but at 30% MVC. Short-term endurance training has no effect on maximal RFD or muscle activation at the onset of contraction [50]. However, endurance training performed in addition to power training can limit the development of explosivity (e.g., [61]), as developed in detail elsewhere in this book.

Training Interventions: Effects on CNS During a Fatiguing Exercise

Endurance training is known to increase performance, yet the time to task failure (or other type of performance index) does not provide information regarding neural or muscular contributions to the improved resistance to fatigue. This last section will only focus on that aspect. One of the first longitudinal studies to investigate the effect of endurance training on neural adaptations was performed by Cafarelli et al. [62]. This study showed that after training, a smaller additional recruitment of MUs occurred during a 20-min cycling exercise compared to pre-training because of attenuated muscle fatigue, in line with the correlation that we found between the change in RMS and muscle fatigue during a cycling exercise [63]. The smaller MU recruitment was accompanied by a reduction in force sensation. Since this initial study, only three very recent (2015–2017) experiments have examined the issue of the change in neuromuscular fatigue after a period of endurance training. A summary of subject characteristics and training protocols for these three studies can be found in Table 4.1. The fatiguing tasks for these studies are described as follows.

Behrens et al. [50] performed a fatiguing task consisting of dynamic plantar flexion in isotonic mode of a Cybex dynamometer at 40% isometric MVC at a frequency of 35 contractions/min. Only fatigue at isotime compared to pre-training was considered post-training, and cycling performance was not measured. In Zghal et al. [53], subjects were asked to sustain a low-intensity isometric muscle action (15% of maximal isometric force) until task failure. Post-training, subjects performed a fatiguing exercise until task failure and one at isotime compared to pre-training. Even more interesting is O'Leary et al. [25], where adaptations were quantified at post-training during cycling trials at both absolute exercise intensity (based on pre-training VO_2max) and relative exercise intensity (based on post-training VO_2max). Importantly, only O'Leary et al. [25] investigated fatigue during a task specific to the training performed. Below, we differentiate the effects of training (1) for a given workload (i.e., at isotime) and (2) at task failure.

1. *For a given workload.* Surprisingly, Behrens et al. [50] showed that although some neuromuscular parameters tended to be improved in the endurance-trained group compared to controls following fatiguing exercise (e.g., isometric MVC, normalized V-wave, and peak twitch torque), none reached statistical significance. However, a comparison of the changes in isometric MVC after the fatiguing protocol before and after the training with a paired t-test indicated a significant improvement in fatigue resistance. The two other protocols found significant reductions of fatigue indices. Both Zghal et al. [53] and O'Leary et al. [25] demonstrated that endurance training could delay the time course of central fatigue. The investigators attributed these findings to the reduction in inhibitory afferent feedback from exercising muscles due to an improved muscle oxidative metabolism, i.e., less peripheral fatigue as demonstrated in the two studies. Importantly, the effects of endurance training on the rapid component of VO_2 kinetics could also play a role in minimizing peripheral fatigue as we recently showed [64]. In addition to the reduction of muscle fatigue after training, it has been suggested that the sensitivity of muscle afferents may be reduced (i.e., up-regulation of their threshold) in response to physical training such that central fatigue may be reduced and/or delayed after training. However, no direct evidence currently supports this hypothesis [53]. Neither of these two investigations found any excitability or inhibitory adaptations of the corticospinal pathway following training but Zghal et al. [53] found that the increase in SP duration with fatigue was abolished at isotime after training (SPs were not measured in [25]). It has also been reported that under fatiguing conditions, motor unit conduction velocity declines less during a fatiguing task (30% MVC to task failure) after 6 weeks of endurance training when compared to pre-training [65].
2. *At task failure.* Peripheral fatigue was greater in both O'Leary et al. [25] and Zghal et al. [53] after training. Interestingly, the greater peripheral fatigue at task failure only occurred after the high-intensity interval training protocol and not after the low-intensity continuous one in O'Leary et al. [25]. Two adaptations within the nervous system could explain the greater peripheral fatigue at task failure after training: (1) the up-regulation in the III-IV afferent firing threshold

suggested above [53] that would result in greater tolerance to metabolic disturbances and translate into an accumulation of greater peripheral fatigue and/or (2) improvement of sensorimotor system tolerance to noxious inhibitory signals, i.e., better pain tolerance (e.g., to acidosis) by the central nervous system. In this regard, one of the authors of this book chapter proposed a “flush model” to describe performance regulation in endurance and ultra-endurance exercise [30]. Improved sensorimotor system tolerance would represent a reduction of the “security reserve” in the brain, which is influenced by acceptable levels of peripheral fatigue for the subject (among other factors).

Contrary to peripheral fatigue, central fatigue at task failure either decreases [25] or is maintained [53] after training. In fact, O’Leary et al. [25] demonstrated that 6 weeks of high-intensity interval training significantly attenuated the exercise-induced reduction in VA_{PNS} . Although these investigators found a significant improvement in time to task failure following both interval and continuous training at absolute exercise intensity, they did not observe any change in corticospinal excitability (MEPs) or inhibition (SPs) or in supraspinal voluntary activation (VA_{TMS}) following the two modes of training. Besides the possible up-regulation of group III-IV afferents, O’Leary et al. [25] speculated that adaptations in central mechanisms (improved handling of serotonin [66], mitigated cerebral ammonia uptake [67], improved cerebral oxygenation [68], and enhanced spinal reflex excitability [44]) may also occur. A limitation of O’Leary et al. [25] is that participants needed to move from the cycle-ergometer to the isometric chair for post-exercise assessments. This process resulted in a time-delay to the neuromuscular assessment which could possibly have reduced the magnitude of corticospinal responses [13] (see section “Tools Used to Assess Neural Adaptations” on methodological considerations).

In summary, central fatigue induced by endurance exercise has been extensively investigated over the last 15 years and has been found to play a major role in neuromuscular fatigue, especially with prolonged running, even if the assessment tools available to date have not fully allowed for complete understanding of fatigue etiology. On the contrary, neural responses to endurance training are surprisingly under-investigated. For instance, only one study has investigated the effects of training on fatigue during a task specific to the type of training performed. Similar to muscle adaptations, most neural changes are specific to endurance training when compared to explosive or strength training. Yet the literature regarding spinal excitability (H-reflex) and cortical excitability/inhibition (TMS parameters) in the unfatigued state is rather uninformative. After training, central fatigue is reduced for a given absolute load, probably related to the reduction in inhibitory afferent feedback since peripheral fatigue is reduced as expected, even if lower sensitivity of group III and IV afferents may also play a role. Although no changes in corticospinal excitability or inhibition or supraspinal voluntary activation were found after exercise to task failure, peripheral fatigue is greater (especially when training includes interval training) but central fatigue is attenuated when measured with traditional nerve stimulation techniques. These training adaptations could benefit endurance/ultra-endurance performance, especially since the longer the event, the greater the amount

of central fatigue that develops. Further longitudinal studies, in particular studies examining changes in neuromuscular function post-training with state-of-the-art techniques, are needed to better understand how the central nervous system adapts to chronic, low-intensity exercise.

References

1. Burtin C, Saey D, Saglam M, Langer D, Gosselink R, Janssens W, Decramer M, Maltais F, Troosters T. Effectiveness of exercise training in patients with COPD: the role of muscle fatigue. *Eur Respir J*. 2012;40:338–44.
2. Gathercole RJ, Stellingwerff T, Sporer BC. Effect of acute fatigue and training adaptation on countermovement jump performance in elite snowboard cross athletes. *J Strength Cond Res*. 2015;29:37–46.
3. Millet GY, Martin V, Martin A, Verges S. Electrical stimulation for testing neuromuscular function: from sport to pathology. *Eur J Appl Physiol*. 2011;111:2489–500.
4. Belanger AY, McComas AJ. Extent of motor unit activation during effort. *J Appl Physiol Respir Environ Exerc Physiol*. 1981;51:1131–5.
5. Todd G, Taylor JL, Gandevia SC. Measurement of voluntary activation of fresh and fatigued human muscles using transcranial magnetic stimulation. *J Physiol*. 2003;551:661–71.
6. McNeil CJ, Butler JE, Taylor JL, Gandevia SC. Testing the excitability of human motoneurons. *Front Hum Neurosci*. 2013;7:152.
7. Taylor JL, Gandevia SC. Noninvasive stimulation of the human corticospinal tract. *J Appl Physiol* (1985). 2004;96:1496–503.
8. Taylor JL, Gandevia SC. Transcranial magnetic stimulation and human muscle fatigue. *Muscle Nerve*. 2001;24:18–29.
9. Yacyshyn AF, Woo EJ, Price MC, McNeil CJ. Motoneuron responsiveness to corticospinal tract stimulation during the silent period induced by transcranial magnetic stimulation. *Exp Brain Res*. 2016;234:3457–63.
10. Temesi J, Rupp T, Martin V, Arnal PJ, Feasson L, Verges S, Millet GY. Central fatigue assessed by transcranial magnetic stimulation in ultratrail running. *Med Sci Sports Exerc*. 2014;46:1166–75.
11. Doyle-Baker D, Temesi J, Medysky ME, Holash RJ, Millet GY. An innovative ergometer to measure neuromuscular fatigue immediately after cycling. *Med Sci Sports Exerc*. 2018;50(2):375–87.
12. Froyd C, Millet GY, Noakes TD. The development of peripheral fatigue and short-term recovery during self-paced high-intensity exercise. *J Physiol*. 2013;591:1339–46.
13. Mira J, Lapole T, Souron R, Messonnier L, Millet GY, Rupp T. Cortical voluntary activation testing methodology impacts central fatigue. *Eur J Appl Physiol*. 2017;117(9):1845–57.
14. Taylor JL, Allen GM, Butler JE, Gandevia SC. Supraspinal fatigue during intermittent maximal voluntary contractions of the human elbow flexors. *J Appl Physiol*. 2000;89:305–13.
15. Millet GY, Martin V, Lattier G, Ballay Y. Mechanisms contributing to knee extensor strength loss after prolonged running exercise. *J Appl Physiol*. 2003;94:193–8.
16. Jubeau M, Rupp T, Perrey S, Temesi J, Wuyam B, Levy P, Verges S, Millet GY. Changes in voluntary activation assessed by transcranial magnetic stimulation during prolonged cycling exercise. *PLoS One*. 2014;9:e89157.
17. Sidhu SK, Bentley DJ, Carroll TJ. Locomotor exercise induces long-lasting impairments in the capacity of the human motor cortex to voluntarily activate knee extensor muscles. *J Appl Physiol*. 2009;106:556–65.
18. Klass M, Roelands B, Levenez M, Fontenelle V, Pattyn N, Meeusen R, Duchateau J. Effects of noradrenaline and dopamine on supraspinal fatigue in well-trained men. *Med Sci Sports Exerc*. 2012;44:2299–308.

19. Lepers R, Maffiuletti NA, Rochette L, Brugniaux J, Millet GY. Neuromuscular fatigue during a long-duration cycling exercise. *J Appl Physiol*. 2002;92:1487–93.
20. Martin V, Kerhervé H, Messonnier LA, Banfi JC, Geyssant A, Bonnefoy R, Féasson L, Millet GY. Central and peripheral contributions to neuromuscular fatigue induced by a 24-h treadmill run. *J Appl Physiol*. 2010;108:1224–33.
21. Millet GY, Tomazin K, Verges S, Vincent C, Bonnefoy R, Boisson RC, Gergele L, Feasson L, Martin V. Neuromuscular consequences of an extreme mountain ultra-marathon. *PLoS One*. 2011;6:e17059.
22. Place N, Lepers R, Deley G, Millet GY. Time course of neuromuscular alterations during a prolonged running exercise. *Med Sci Sports Exerc*. 2004;36:1347–56.
23. Temesi J, Arnal PJ, Rupp T, Feasson L, Cartier R, Gergele L, Verges S, Martin V, Millet GY. Are females more resistant to extreme neuromuscular fatigue? *Med Sci Sports Exerc*. 2015;47:1372–82.
24. Sidhu SK, Cresswell AG, Carroll TJ. Motor cortex excitability does not increase during sustained cycling exercise to volitional exhaustion. *J Appl Physiol* (1985). 2012;113:401–9.
25. O’Leary TJ, Collett J, Howells K, Morris MG. Endurance capacity and neuromuscular fatigue following high- vs moderate-intensity endurance training: a randomized trial. *Scand J Med Sci Sports*. 2017;27(12):1648–61.
26. Racinais S, Girard O, Micallef JP, Perrey S. Failed excitability of spinal motoneurons induced by prolonged running exercise. *J Neurophysiol*. 2007;97:596–603.
27. Sidhu SK, Weavil JC, Mangum TS, Jessop JE, Richardson RS, Morgan DE, Amann M. Group III/IV locomotor muscle afferents alter motor cortical and corticospinal excitability and promote central fatigue during cycling exercise. *Clin Neurophysiol*. 2017;128:44–55.
28. Weavil JC, Sidhu SK, Mangum TS, Richardson RS, Amann M. Fatigue diminishes motoneuronal excitability during cycling exercise. *J Neurophysiol*. 2016;116:1743–51.
29. Goodall S, Gonzalez-Alonso J, Ali L, Ross EZ, Romer LM. Supraspinal fatigue after normoxic and hypoxic exercise in humans. *J Physiol*. 2012;590:2767–82.
30. Millet GY. Can neuromuscular fatigue explain running strategies and performance in ultra-marathons?: the flush model. *Sports Med*. 2011;41:489–506.
31. Millet GY, Lepers R. Alterations of neuromuscular function after prolonged running, cycling and skiing exercises. *Sports Med*. 2004;34:105–16.
32. Millet GY, Martin V, Maffiuletti NA, Martin A. Neuromuscular fatigue after a ski skating marathon. *Can J Appl Physiol*. 2003;28:434–45.
33. Millet GY, Lepers R, Maffiuletti NA, Babault N, Martin V, Lattier G. Alterations of neuromuscular function after an ultramarathon. *J Appl Physiol*. 2002;92:486–92.
34. Avela J, Kyrolainen H, Komi P. Altered reflex sensitivity after repeated and prolonged passive muscle stretching. *J Appl Physiol*. 1999;86:1283–91.
35. Coffey VG, Hawley JA. The molecular bases of training adaptation. *Sports Med*. 2007;37:737–63.
36. Egan B, Zierath JR. Exercise metabolism and the molecular regulation of skeletal muscle adaptation. *Cell Metab*. 2013;17:162–84.
37. Aagaard P, Simonsen EB, Andersen JL, Magnusson P, Dyhre-Poulsen P. Neural adaptation to resistance training: changes in evoked V-wave and H-reflex responses. *J Appl Physiol* (1985). 2002;92:2309–18.
38. Carroll TJ, Riek S, Carson RG. The sites of neural adaptation induced by resistance training in humans. *J Physiol*. 2002;544:641–52.
39. Van Cutsem M, Duchateau J, Hainaut K. Changes in single motor unit behaviour contribute to the increase in contraction speed after dynamic training in humans. *J Physiol*. 1998;513(Pt 1):295–305.
40. Lattier G, Millet GY, Maffiuletti NA, Babault N, Lepers R. Neuromuscular differences between endurance-trained, power-trained, and sedentary subjects. *J Strength Cond Res*. 2003;17:514–21.
41. Garrandes F, Colson SS, Pensini M, Seynnes O, Legros P. Neuromuscular fatigue profile in endurance-trained and power-trained athletes. *Med Sci Sports Exerc*. 2007;39:149–58.

42. Cohen R, Mitchell C, Dotan R, Gabriel D, Klentrou P, Falk B. Do neuromuscular adaptations occur in endurance-trained boys and men? *Appl Physiol Nutr Metab*. 2010;35:471–9.
43. Hakkinen K, Keskinen KL. Muscle cross-sectional area and voluntary force production characteristics in elite strength- and endurance-trained athletes and sprinters. *Eur J Appl Physiol Occup Physiol*. 1989;59:215–20.
44. Vila-Cha C, Falla D, Correia MV, Farina D. Changes in H reflex and V wave following short-term endurance and strength training. *J Appl Physiol* (1985). 2012;112:54–63.
45. Maffiuletti NA, Martin A, Babault N, Pensini M, Lucas B, Schieppati M. Electrical and mechanical H(max)-to-M(max) ratio in power- and endurance-trained athletes. *J Appl Physiol* (1985). 2001;90:3–9.
46. Kocaja DM, Davison E, Robertson CT. Neuromuscular characteristics of endurance- and power-trained athletes. *Res Q Exerc Sport*. 2004;75:23–30.
47. Earles DR, Dierking JT, Robertson CT, Kocaja DM. Pre- and post-synaptic control of moto-neuron excitability in athletes. *Med Sci Sports Exerc*. 2002;34:1766–72.
48. Kumpulainen S, Avela J, Gruber M, Bergmann J, Voigt M, Linnamo V, Mrachacz-Kersting N. Differential modulation of motor cortex plasticity in skill- and endurance-trained athletes. *Eur J Appl Physiol*. 2015;115:1107–15.
49. Kyrolainen H, Komi PV. Stretch reflex responses following mechanical stimulation in power- and endurance-trained athletes. *Int J Sports Med*. 1994;15:290–4.
50. Behrens M, Weippert M, Wassermann F, Bader R, Bruhn S, Mau-Moeller A. Neuromuscular function and fatigue resistance of the plantar flexors following short-term cycling endurance training. *Front Physiol*. 2015;6:145.
51. Perot C, Goubel F, Mora I. Quantification of T- and H-responses before and after a period of endurance training. *Eur J Appl Physiol Occup Physiol*. 1991;63:368–75.
52. Vila-Cha C, Falla D, Farina D. Motor unit behavior during submaximal contractions following six weeks of either endurance or strength training. *J Appl Physiol* (1985). 2010;109:1455–66.
53. Zghal F, Cottin F, Kenoun I, Rebai H, Moalla W, Dogui M, Tabka Z, Martin V. Improved tolerance of peripheral fatigue by the central nervous system after endurance training. *Eur J Appl Physiol*. 2015;115:1401–15.
54. Zghal F, Martin V, Thorkani A, Arnal PJ, Tabka Z, Cottin F. Effects of endurance training on the maximal voluntary activation level of the knee extensor muscles. *Eur J Appl Physiol*. 2014;114:683–93.
55. de Haan A, Gerrits KH, de Ruiter CJ. Counterpoint: the interpolated twitch does not provide a valid measure of the voluntary activation of muscle. *J Appl Physiol* (1985). 2009;107:355–7; discussion 357–8.
56. Shield A, Zhou S. Assessing voluntary muscle activation with the twitch interpolation technique. *Sports Med*. 2004;34:253–67.
57. Jensen JL, Marstrand PC, Nielsen JB. Motor skill training and strength training are associated with different plastic changes in the central nervous system. *J Appl Physiol* (1985). 2005;99:1558–68.
58. Goodwill AM, Pearce AJ, Kidgell DJ. Corticomotor plasticity following unilateral strength training. *Muscle Nerve*. 2012;46:384–93.
59. Griffin L, Cafarelli E. Transcranial magnetic stimulation during resistance training of the tibialis anterior muscle. *J Electromyogr Kinesiol*. 2007;17:446–52.
60. Weier AT, Pearce AJ, Kidgell DJ. Strength training reduces intracortical inhibition. *Acta Physiol (Oxf)*. 2012;206:109–19.
61. Häkkinen K, Alen M, Kraemer WJ. Neuromuscular adaptations during concurrent strength and endurance versus strength training. *Eur J Appl Physiol*. 2003;89:42–52.
62. Cafarelli E, Liebesman J, Kroon J. Effect of endurance training on muscle activation and force sensation. *Can J Physiol Pharmacol*. 1995;73:1765–73.
63. Decorte N, Lafaix PA, Millet GY, Wuyam B, Verges S. Central and peripheral fatigue kinetics during exhaustive constant-load cycling. *Scand J Med Sci Sports*. 2012;22:381–91.

64. Temesi J, Mattioni Maturana F, Peyrard A, Piuco T, Murias JM, Millet GY. The relationship between oxygen uptake kinetics and neuromuscular fatigue in high-intensity cycling exercise. *Eur J Appl Physiol.* 2017;117:969–78.
65. Vila-Cha C, Falla D, Correia MV, Farina D. Adjustments in motor unit properties during fatiguing contractions after training. *Med Sci Sports Exerc.* 2012;44:616–24.
66. Jakeman PM, Hawthorne JE, Maxwell SR, Kendall MJ, Holder G. Evidence for downregulation of hypothalamic 5-hydroxytryptamine receptor function in endurance-trained athletes. *Exp Physiol.* 1994;79:461–4.
67. Nybo L, Dalsgaard MK, Steensberg A, Moller K, Secher NH. Cerebral ammonia uptake and accumulation during prolonged exercise in humans. *J Physiol.* 2005;563:285–90.
68. Rooks CR, Thom NJ, McCully KK, Dishman RK. Effects of incremental exercise on cerebral oxygenation measured by near-infrared spectroscopy: a systematic review. *Prog Neurobiol.* 2010;92:134–50.